

## A Response to Steven Durlauf and James Heckman

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Science makes progress with rational debate, disagreement, and discussion. In that vein, I am grateful to the authors of the comment, Steven Durlauf and James Heckman, for engaging in one of the most important social issues of our time.

Their comment can be synthesized into three points. First, Durlauf and Heckman believe we make claims about discrimination, or the absence thereof, when no such conclusions using a direct regression approach are justified. Second, they describe issues with the endogeneity of police-civilian contacts and suggest modelling those interactions. And, third, they express skepticism of any analysis that relies on police narratives regarding interactions with civilians.

Below, I respond to each of these in turn. Here is a brief summary: I disagree with their first point -- we went through painstaking effort to belabor what our analysis can and cannot teach us about race and policing. We use the term “racial differences” 114 times in lieu of the more prescriptive wording -- “racial discrimination”. We use the phrase “conditional on an interaction” 20 times throughout the paper and explicitly note at the outset -- 5th paragraph -- that if we don’t use this phrase, it’s implied. I’m not sure how many more ways we would have needed to caveat our results to satisfy the authors.

I agree with their second point, theoretically -- that’s why the section immediately following the data description is entitled, “A Note on Potential Selection into Police Data Sets” -- but I’m perplexed as to why they claim our sample is “based on stops.” The truth is the vast majority of the data they are referring to is gleaned from 911 calls for service in which a civilian requests police presence, not police lurking to harass civilians. I agree that police shootings that begin with police stopping a civilian and that interaction escalating and becoming deadly is deeply problematic in terms of understanding the role of race, even conditional upon an interaction. It’s simply not how most officer-involved shootings begin in the actual data. The types of interactions they describe -- “based on stops” -- comprise less than 10% of our sample. Their third point is an important one, but we cannot just throw up our hands because most data available to researchers are collected by the police.

## False Conclusions?

Durlauf and Heckman write, “*The absence of racial differences is treated as sufficient evidence for not further exploring the presence of discrimination, just as the presence of such differences is treated as prima facie evidence of discrimination in the use of force at lesser levels of police response than shooting.*”

I fear the authors did not read the paper carefully. Here are five of many examples that they overlooked:

- Page 1215: “...given the inability to randomly assign race, one can never be confident in the direct regression approach when interpreting racial disparities.”
- Page 1216: “Taken together, we argue that the results are most consistent with, but in no way proof of, taste-based discrimination among police officers who face convex costs of excessive use of force. Yet the data do more to provide a more compelling case that there is no discrimination in officer-involved shootings than they do to illuminate the reasons behind racial differences in nonlethal uses of force.”
- Page 1222: “The OIS data have several notable limitations... Fourth, it is impossible to capture all variables of importance at the time of a shooting. Thus, what appears to be discrimination to some may look like mismeasured contextual factors to others.”
- Page 1239: “Putting the results from the Stop and Frisk and PPCS data sets together, a pattern emerges. Relative to whites, blacks and Hispanics seem to have very different interactions with law enforcement—interactions that are consistent with, though definitely not proof of, some form of discrimination.”
- Page 1258: “In the end, however, without randomly assigning race, we have no definitive proof of discrimination. ”

Importantly, our statistical methods are not new. Our way of describing racial disparities in outcomes is no different than the vast literature in economics spanning more than 70 years describing racial disparities on outcomes such as test scores, health, and wages. Neal and Johnson (1996) is a good example. Ironically, in commenting on that paper, which uses similar methods, Heckman (1998) remarked, “Their findings are important for interpreting the sources of black-white disparity in labor market outcomes.”

## Endogeneity of Police-Civilian Interactions

Thinking through the endogeneity of police-civilian interactions is important. Again, we tried to make this clear in the paper:

- Page 1212: “Unless otherwise noted, all results are conditional on an interaction. Understanding potential selection into police data sets due to bias in whom police interact with is a difficult endeavor. Section III attempts to help get a sense of potential bias in police interactions. Put simply, if one assumes that police simply stop whomever they want for no particular reason, there seem to be large racial differences. If one assumes they are trying to prevent violent crimes, then evidence for bias is small.”
- Page 1244: “To be clear, the empirical thought experiment here is that a police officer arrives at a scene and decides whether or not to use lethal force. Our estimates suggest that this decision is not correlated with the race of the suspect. This does not, however, rule out the possibility that there are important racial differences in whether or not these police-civilian interactions occur at all.”

And, we went well beyond simply providing these caveats to our analysis and tried to better understand how endogeneity of police-civilian contact might affect the results. In the original paper, we coded variables characterizing the reason the officer was at the scene -- from Responding to a Homicide (which is seemingly more exogenous) - coded as 1 if the officers were responding to a violent crime and 0 otherwise to Responding to Suspicious Activity (which is seemingly more endogenous) - coded as 1 if the officers were responding to a suspect engaging in suspicious activity and 0 otherwise.

Durlauf and Heckman seem to assume that the modal police shooting germinates from the types of interactions we have witnessed in the majority of the 13 incidents that have garnered national attention -- police officers selectively interact with a Black civilian and it, for a variety of reasons, escalates into a deadly encounter.<sup>1</sup> This is significantly different from what we see in the data. The vast majority of officer-involved shootings in our data begin with a 911 call for service such as robbery in progress, violent crime in progress, or suicide in progress, not a traffic stop for a broken taillight.

In Houston, when 911 calls are received, the closest available officer is expected to respond if possible -- as their explicit objective is to minimize response times. In these cases,

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<sup>1</sup>Of the 13 incidents that have captured public attention, 7 start with a stop that escalated into a shooting. Two of them began as 911 calls about auto crime, 2 about suspicious activity, and 2 about robbery.

the endogeneity of police-civilian contact is less worrisome than an officer patrolling for suspicious activities. This suggests partitioning the set of officer-involved shootings by their origin, some will originate from 911 calls for service for violent crimes and others will originate because of more endogenous activities like street stops. If the endogeneity of police-civilian interactions is masking important racial disparities in officer-involved shootings, one might expect to find racial disparities on the set of shootings that are more exogenous and no racial disparities on those that are more endogenous.

In contrast, our results demonstrate no racial disparities in officer-involved shootings, independent of the origin of the interaction. As we state in the original paper:

- Page 1247: “Estimated race coefficients across call types—whether officers were dispatched because of a violent crime, robbery, auto crime, or other type of call—are all negative, if anything. This is particularly interesting in light of the potential selection into the sample of OIS cases discussed earlier. Indeed, the majority of police shootings in our data occur during violent crimes or robberies, and on these call types, blacks are less likely to be shot, if anything.”

It is important to note, however, that these results do not rule out the possibility that police are “strategically biased” -- choosing not to have their actions correlate with race in more exogenous interactions, such as 911 calls, but allowing race to affect their decisions in more endogenous interactions. In this sense, it is not about whether cops are biased or not, it is more about when they can apply their bias. If this is the case, the “no disparity” result would still be meaningful in the set of more exogenous interactions and have less meaning in the other 9% of the sample.

#### Reliance on Police Narratives

All OIS narratives are gleaned from police descriptions so as to be able to code data around the context of the shooting. We understood that this was a potential issue and described it clearly in the paper:

- Page 1222: “The OIS data have several notable limitations. Taken alone, officer involved shootings are the most extreme and least used form of police force and thus, in isolation, may be misleading. Second, the penalties for wrongfully discharging a lethal weapon in any given situation can be life altering; thus, the incentive to misrepresent contextual factors on police reports may be large. Third, we do not typically have the suspect’s side of the story, and often there are no witnesses.”

Our analysis does not rely on conditioning variables to obtain the results, but this only partially addresses their concern. Our main risk set is the group of arrestees for which police could have justifiably used force -- according to departmental standards -- but chose not to. An important concern is that there may be racial bias in the determination of who is in this risk set. In an extreme example to illustrate the point -- if a black civilian walking down the street has the same probability of ending up among those arrestees as a white civilian wielding a knife, then our estimates are biased and the fact that we find the same results without controls does not matter because the risk set itself is compromised.

This is difficult, if not impossible, to overcome. In an effort to partially get at it, we displayed results by year. To the extent that there was less public or private scrutiny on police officers early in our sample -- when video-enabled smartphones were less present relative to later in our sample when Black Lives Matter became prominent -- there may have been less incentive to lie and thus the bias in the risk set may be different over time. For instance, if we found racial disparities in the early 2000s but none in the later years of our sample, one might be worried that reporting bias might explain the later results. Interestingly, racial differences in OIS between 2000 and 2015 are remarkably constant.

As things stand today, police narratives are the only source that contain detailed information about any shooting event. If we dislike dependence on police narratives, then we must find other innovative ways to get information. We would all benefit from such data.